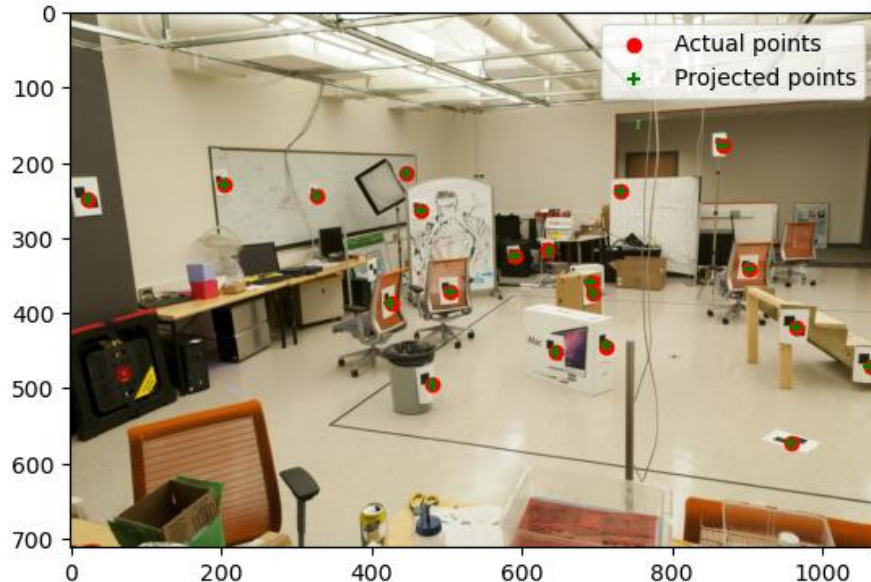


CS 4476 PS 3

Devin Fromond
dfromond3@gatech.edu
903761713

Part 1.5: Projection Matrix for provided image

<insert visualization of projected 3D points and actual 2D points for image provided by us here [1]>

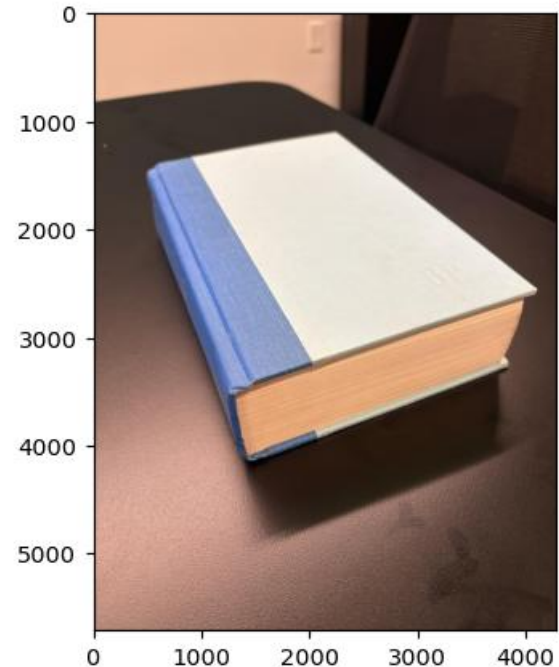
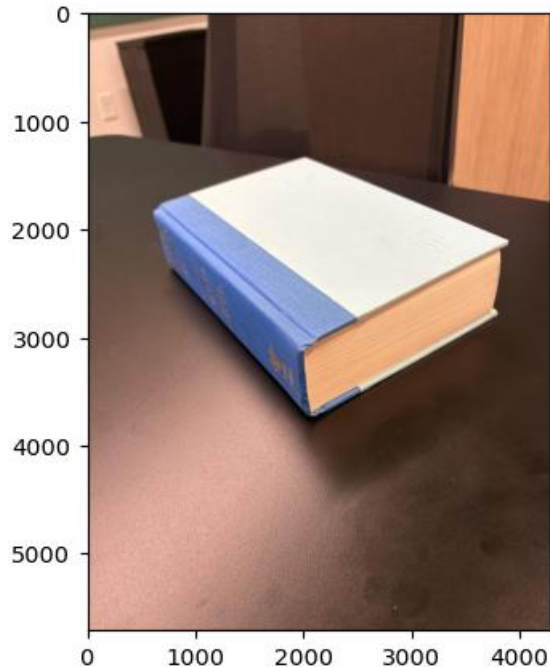


<What is the minimum number of 3D-2D point correspondences needed to estimate the projection matrix? Why? [2]>

The minimum number of correspondences is 6 since the projection matrix has 11 independent parameters and each pair of 3D-2D mappings provide two equations. 2 times 6 is 12 which is greater than or equal to 11. We cannot do 5 since this would only correspond to 10 independent parameters.

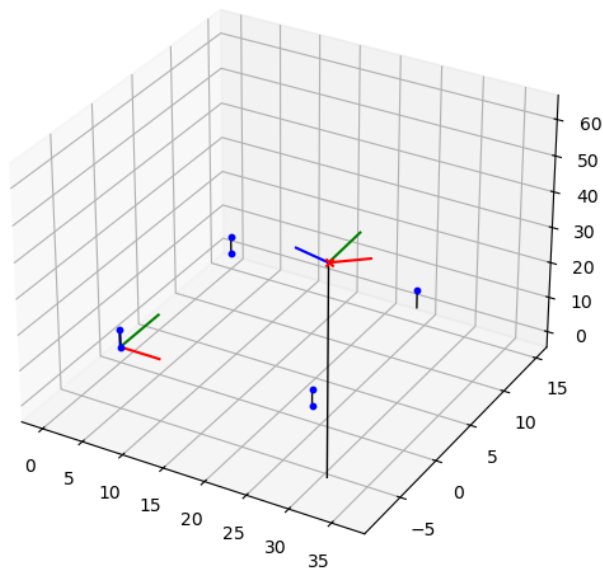
Part 2.1: Projection Matrix for custom images

<Copy two images of your fiducial object here [2]>

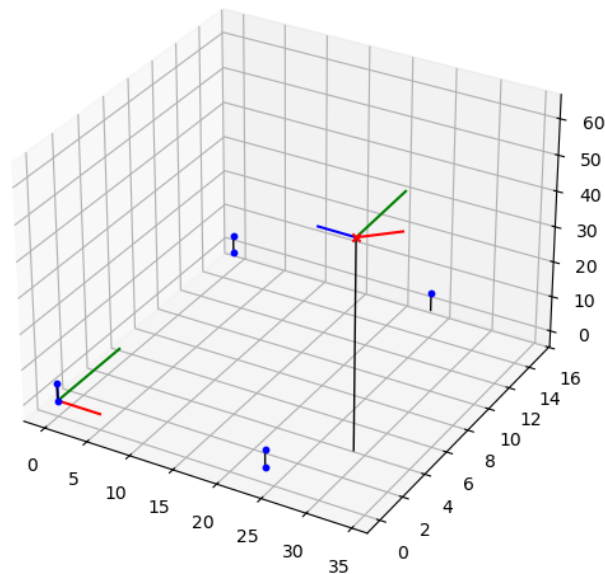


Part 2.2: Pose init for custom images

<Insert visualization for the initialized camera pose for 1st image> [1]

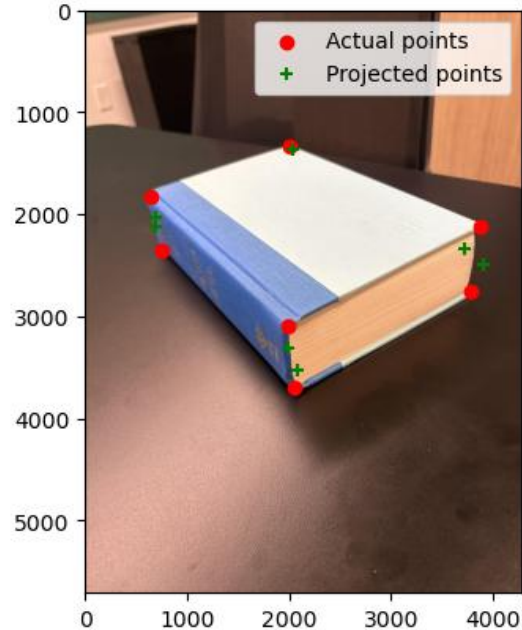


<Insert visualization for the initialized camera pose for 2nd image> [1]

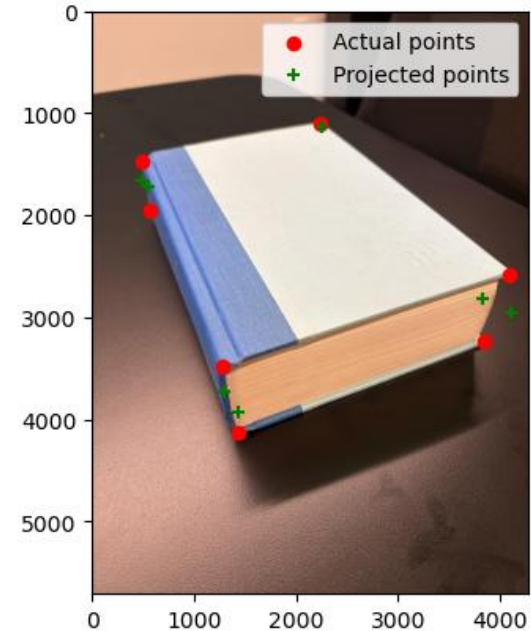


Part 2.2: Optimized results for custom images

<Insert visualization for projected 3D points and actual 2D points for 1st image> [1.5]

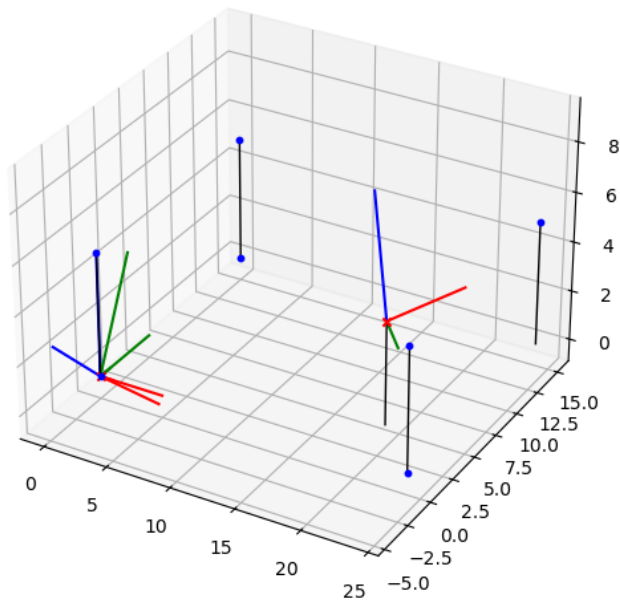


<Insert visualization for projected 3D points and actual 2D points for 2nd image> [1.5]



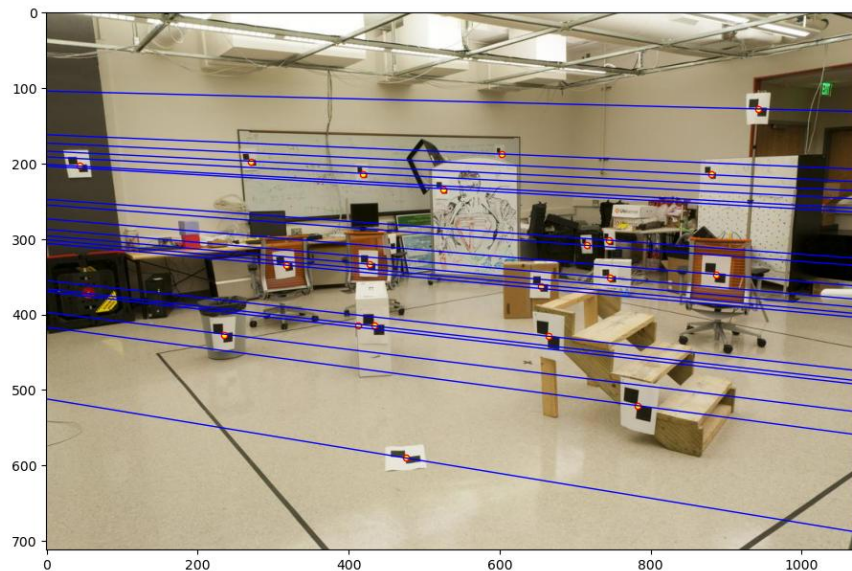
Part 2.3: Optimized Camera Poses

<Insert pose with world and optimized camera's coordinate systems [1]>

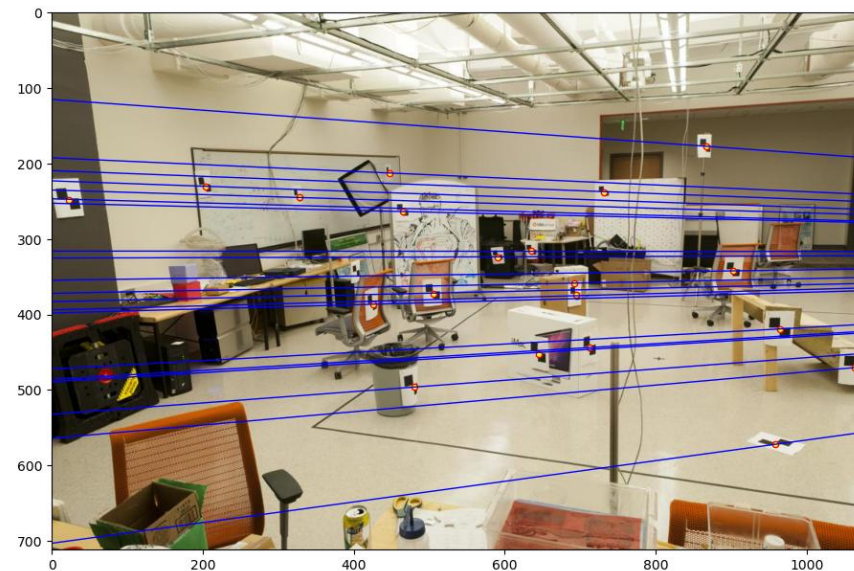


Part 3.2: Optimized Epipolar Lines (given images)

<Insert left image with epipolar lines> [1]

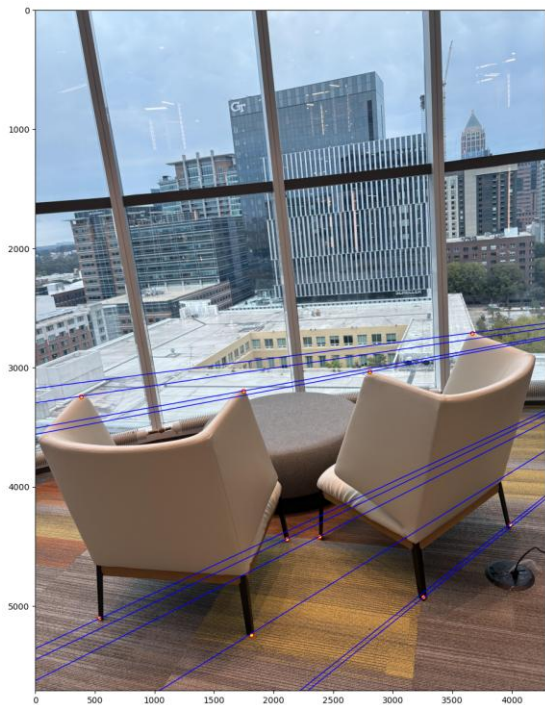


<Insert right image with epipolar lines> [1]

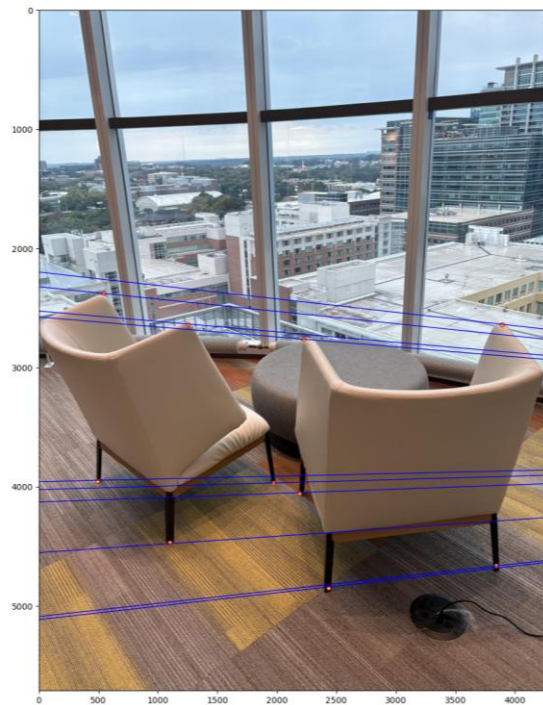


Part 3.3: Optimized Epipolar Lines (custom images)

<Insert left image with epipolar lines> [1.5]



<Insert right image with epipolar lines> [1.5]



Part 3.4: Reflection Questions [1x3]

1. Rotating the camera alone does not change the camera center. The fundamental matrix cannot be calculated as a result.
2. From the lectures, we know the fundamental matrix maps a point to an epipolar line (not a point), so we are unable to determine exact position without depth information.
3. If the camera centers are visible, then the epipoles occur inside the image and epipolar lines radiate from them.

Part 3.4: Reflection Questions [1x3]

4. If all of the lines are horizontal, then the corresponding points lie on the same row – this also means the images are rectified.
5. The fundamental matrix is defined up to scale since the equation $x'^T F x = 0$ is homogeneous.
6. The fundamental matrix has rank 2 since it is derived from the cross product matrix $[t] \times R$ – which is necessarily singular.

Part 4.2: RANSAC Iterations Questions [1x3]

Type your answers to the three RANSAC Iterations questions from the jupyter notebook below:

1. $p=0.999$, $w=0.9$, $s=9$ Iterations = $N = 9$

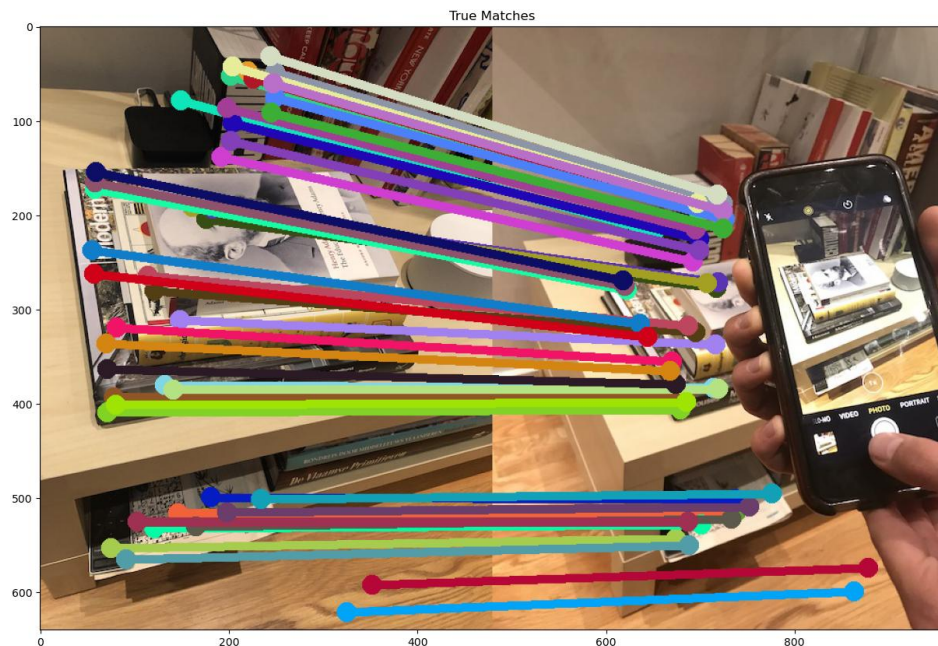
1. $p=0.999$, $w=0.9$, $s=18$ Iterations = $N = 43$

1. $p=0.999$, $w=0.7$, $s=7$ Iterations = $N = 81$

$$N = \left\lceil \frac{\ln(1 - p)}{\ln(1 - w^s)} \right\rceil$$

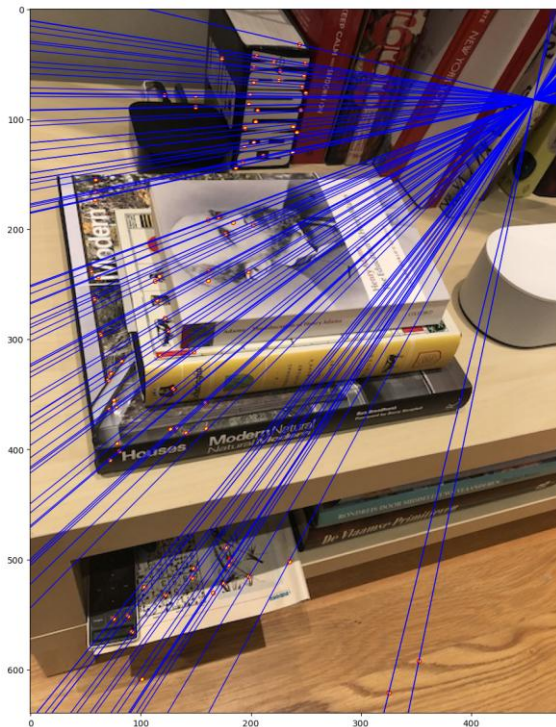
Part 4.4: RANSAC Inlier Matches

<Paste the inlier matches found by Ransac [2]>

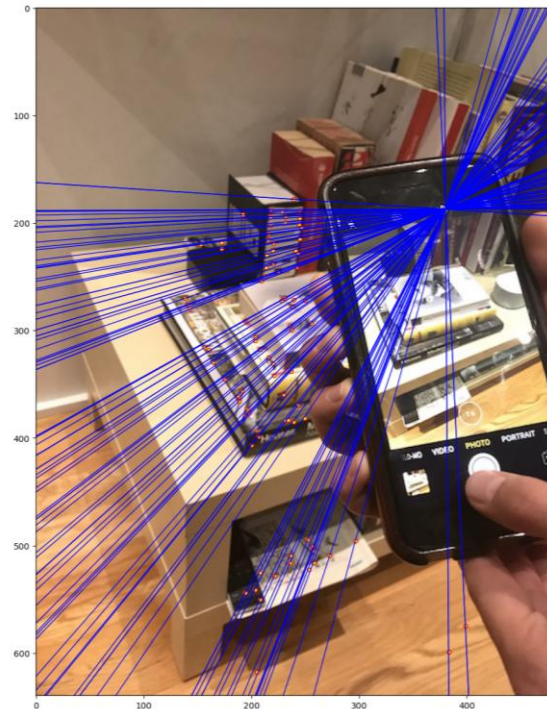


Part 4.4: RANSAC Epipolar Lines

<Paste the left image with epipolar lines> [1]



<Paste the right image with epipolar lines> [1]



Local Unit tests results

<Paste the screenshot when you run all provided unit tests using `pytest`> [1]

```
(ps3) C:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4476\ps3_release\ps3
_unit_tests>pytest ./
===== test session starts =====
platform win32 -- Python 3.9.24, pytest-8.4.2, pluggy-1.5.0
rootdir: C:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4476\ps3_release
plugins: anyio-4.10.0
collected 20 items

part1_unit_test.py ..... [ 25%]
test_essential_matrix_decomposition.py .. [ 35%]
test_fundamental_matrix.py ..... [ 80%]
test_ransac.py .... [100%]

===== 20 passed in 34.70s =====

(ps3) C:\Users\devin\OneDrive\Documents\Georgia_Tech\CS 4476\ps3_release\ps3
_unit_tests>
```

Conclusions

<Describe what you have learned in this project. Feel free to include any challenges you ran into.> [2]

I think that in lecture, the examples were “pretty examples”; it is easy to see which direction to go when it is in front of you. For these examples – ESPECIALLY PART TWO – I struggled (especially with rotation matrix). I have learned how to break apart the matrices and I struggled doing so.